eSC Energy Usage:
How To Reduce Electric Grid Demand

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IST 687: Introduction To Data Science

Syracuse University

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Introduction

With increasing temperatures comes increasing demand for electricity. People will try to cool down in various ways from jumping in pools, blasting their central air conditioning, turing their fans on or even standing infront of the freezer with the door open. All of these heat fighting tactics have one thing in common: they all require energy. We see this year over year, as summer begins, people want to fight the heat but what does that mean for you at eSC? As demand peaks it puts an extreme strain on the power grid, leading a blackout. This is the worst case scenario for eSC and all of its clients in South and parts of North Carolina. Who wants their power turned off in the middle of July? We certainly do not.

For our clients, we lose their trust. They pay on time every month and we ensure we keep that power turned on. We have been hired by eSC to conduct a study based on their existing customer usage data to predict the demand for next summer's "hottest" month, July. It is believed that next summer will be "extra hot" leading to an increase in demand from all customers, placing an immense strain on the grid. Currently, it is unfeasible to build a new energy production facility. Due to this, this report will discuss various approaches we have taken to correlate user behavior to energy usage throughout the month of July. Once we had determined our optimal model, we began conducting predictive analysis to understand how high the demand would be and what eSC can do to mitigate this. At the end of this report, we will provide our suggestions on what is driving the demand on the electric grid and what we can do to reduce it in preparations for next year.

Throughout this project we took two different approaches to arrive at our conclusions. Each one will be submitted and labelled "Approach 1" and "Approach 2". The reason for this is to try different approaches and come to the same or different conclusions based on the findings.

Descriptive Statistics

The data we received from eSC was in three separate files (or sets of files): the first file contains basic house information for over 5,000 homes served by eSC. This is essentially a list of house details that do not change, such as its building ID, what materials it's made of, and its size. The second set of files consists of the energy usage data for each house served by eSC. Each column is a calibrated and validated energy usage amount, derived from 1-hour load profiles, for each house. These columns describe how much energy is used from different sources within the house, such as air conditioning, pool heaters, and laundry machines. The third set of files consists of hour-by-hour weather information within each county. Within this file there is a minimum 1 of the 5,000 houses. Each house in our data is spread across 50 counties. The basic house information file and energy usage files were both given to us in 'parquet' format, which is a CSV file that has been optimized for storage. The weather information was given to us as a set of CSV files. We were also provided with a CSV consisting of the metadata for each attribute across the energy usage

data and house information data. While we did not incorporate this file into the code, it was a helpful guide that allowed us to gain a better understanding of what we were examining.

Data Acquisition, Cleaning, Transformation, and Munging

The following outlines “Approach 1”, “Approach 2” followed very closely to this methodology with minor differences outlined at the bottom:

We started by creating 3 separate functions. Each output of these functions consists of separate data frames detailing of the given house information, energy usage, and/or weather information for each house and the county it’s located in. For each of these functions, we made sure to subset the output to only show data from the month of July. We started by reading in the basic house information using the read_parquet() function from the arrow package. We then began reading in the weather data. This was the easiest to code because each file was a CSV, so we could just take a given county as input, paste that onto the provided URL for the entire folder, then use the read_csv() function from the tidyverse package to read the pasted URL in as a CSV file. The second function we made was for the energy usage of a given house. We started by reading in the data for the given house ID by pasting the ID onto the overall folder’s URL and using the read_parquet() function to read it in. We then began adding a new column, ‘total_usage’, which was a numeric sum of a given house’s values for every column concerning an energy output. We then proceeded to get rid of every column from the database we felt we did not need anymore (including every column that contributed to the total energy usage) and would prove redundant when creating a predictive model. We also decided to subset the data further to exclude any NA values. The third function we created merged the energy usage data and weather data together. We selected rows from the house information to merge with everything as well. We then used the bind_cols() function in base R to merge everything together. We then created a brand new function that not only took a sample of 250 houses but also merged any sampled rows by their unique building ID and then combined all the merged rows together into one, workable dataframe called mergedDF.

Once we had our large dataframe, we then proceeded to turn all the columns concerning insulation, window material and areas, air conditioning efficiency, and energy usage level into factored values based on each column’s possible set of values, then converted those factor values into numeric values. This made the more string-based columns (those that contain character values) easier to work with so they can factor better into our predictions. We then added one final column to the dataframe; usage_per_sqft. This value detailed how much a house uses based on square footage (if two houses were using the same amount of energy and one house is half as big as the other, it would not make sense to just use the total energy usage variable as a proper assessment of whether each house is using too much energy). This value is derived by taking the total energy usage of a

given house and dividing it by its square footage. It is this value that we will be predicting in our analysis.

We also thought it would be best to create a brand new copy of mergedDF and store it in a new dataframe, modelDF, that we would use for modeling. We then took out several more columns from modelDF that we felt would not contribute much to the linear model. We took out the building and county IDs solely because they are mere location indicators that leave no effect on how much energy is used. For this reason, we also took out the latitude and longitude of the house's location. We also found that the columns indicating roof and rim joist insulation had zero variances and singularities that lowered the accuracy, so in order to reduce errors, we removed those columns as well.

"Approach 2" follows the same methodology as approach one. The key differences are when the files are being merged together, the function process_sample() is called. This function will first download the weather data and combine it with all the counties. Then it will download the energy data for each house based on the building id and then will begin joining the county and energy data. Once that is merged, it will merge the static house data at the end. This reduced the time for merging by a significant factor when compared to "Approach 1". This allowed for the stratified sample to be much larger.

Use of Modeling Techniques and Visualizations

Approach 1

We figured that, since usage_per_sqft is technically a continuous variable, it would be best to use a linear model. Using every remaining column in modelDF as predictors, we ran the linear model. Right off the bat, we found that most of the predictors, such as square footage, temperature, humidity, and most areas of insulation, were statistically significant in predicting the usage per square foot. Admittedly, we weren't able to get this model to an adjusted R-squared higher than around 0.50-0.80 despite trying everything we could to raise it. We got rid of total_usage and every other column that contributes to it, yet whatever we did couldn't raise it for this model. Weirdly enough, when trying with a smaller sample size of 25, we got an adjusted R-squared of around 0.6, though we think a bigger sample would be much more fitting, so we decided not to go with that. However, given that this is still technically the highest accuracy we were able to get, this is the model we'll be using.

```
Call:
lm(formula = usage_per_sqft ~ ., data = modelDF)

Residuals:
      Min        1Q    Median        3Q       Max
-0.0044263 -0.0003547 -0.0000773  0.0002580  0.0174839

Coefficients:
                                                          Estimate Std. Error  t value Pr(>|t|)
(Intercept)                                             -1.967e-02  2.951e-03   -6.665 2.65e-11 ***
in.sqft                                                 -1.990e-07  1.740e-09 -114.351  < 2e-16 ***
in.geometry_stories                                      3.942e-05  3.558e-06   11.077  < 2e-16 ***
in.bedrooms                                             -6.013e-05  2.401e-06  -25.044  < 2e-16 ***
in.hvac_cooling_efficiencyAC, SEER 13                    2.488e-04  5.568e-06   44.688  < 2e-16 ***
in.hvac_cooling_efficiencyAC, SEER 15                    3.687e-04  6.032e-06   61.114  < 2e-16 ***
in.hvac_cooling_efficiencyAC, SEER 8                    -8.849e-05  1.013e-05   -8.733  < 2e-16 ***
in.hvac_cooling_efficiencyHeat Pump                      1.115e-04  5.704e-06   19.547  < 2e-16 ***
in.hvac_cooling_efficiencyNone                           4.454e-04  1.661e-05   26.822  < 2e-16 ***
in.hvac_cooling_efficiencyRoom AC, EER 10.7              -3.995e-05  1.511e-05   -2.644 0.008200 **
in.hvac_cooling_efficiencyRoom AC, EER 12.0              2.405e-04  1.927e-05   12.481  < 2e-16 ***
in.hvac_cooling_efficiencyRoom AC, EER 9.8               8.454e-05  1.811e-05    4.669 3.04e-06 ***
in.usage_levelLow                                       -5.407e-04  4.842e-06 -111.680  < 2e-16 ***
in.usage_levelMedium                                    -5.039e-04  3.938e-06 -127.932  < 2e-16 ***
in.windowsDouble, Clear, Non-metal, Air                 -4.088e-05  5.405e-06   -7.563 3.96e-14 ***
in.windowsDouble, Clear, Non-metal, Air, Exterior Clear Storm  3.160e-04  1.495e-05   21.136  < 2e-16 ***
in.windowsDouble, Low-E, Non-metal, Air, M-Gain          1.344e-04  4.620e-06   29.080  < 2e-16 ***
in.windowsSingle, Clear, Metal                           1.130e-05  5.307e-06    2.129 0.033244 *
in.windowsSingle, Clear, Metal, Exterior Clear Storm     2.665e-05  3.077e-05    0.866 0.386342
in.windowsSingle, Clear, Non-metal                       1.223e-04  5.501e-06   22.238  < 2e-16 ***
in.windowsSingle, Clear, Non-metal, Exterior Clear Storm  2.506e-05  1.464e-05    1.712 0.086983 .
in.windowsTriple, Low-E, Non-metal, Air, L-Gain         -4.509e-04  1.822e-05  -24.748  < 2e-16 ***
in.window_areasF15 B15 L15 R15                          -1.425e-04  5.559e-06  -25.640  < 2e-16 ***
in.window_areasF18 B18 L18 R18                           2.867e-05  4.833e-06    5.932 3.00e-09 ***
in.window_areasF30 B30 L30 R30                           1.111e-04  1.008e-05   11.025  < 2e-16 ***
in.window_areasF6 B6 L6 R6                               2.433e-05  6.793e-06    3.582 0.000341 ***
in.window_areasF9 B9 L9 R9                              -2.560e-05  4.251e-06   -6.022 1.72e-09 ***
in.insulation_ceilingR-13                               -1.803e-04  1.375e-05  -13.106  < 2e-16 ***
in.insulation_ceilingR-19                               -4.794e-05  1.375e-05   -3.486 0.000491 ***
in.insulation_ceilingR-30                               -1.352e-04  1.351e-05  -10.003  < 2e-16 ***
in.insulation_ceilingR-38                                2.474e-05  1.392e-05    1.777 0.075610 .
in.insulation_ceilingR-49                               -1.568e-04  2.894e-05   -5.417 6.05e-08 ***
in.insulation_ceilingR-7                                -2.386e-04  1.479e-05  -16.127  < 2e-16 ***
in.insulation_ceilingUninsulated                        -1.495e-04  1.461e-05  -10.236  < 2e-16 ***
in.insulation_floorCeiling R-19                          1.709e-05  8.267e-06    2.067 0.038739 *
in.insulation_floorNone                                  1.669e-04  7.938e-06   21.020  < 2e-16 ***
in.insulation_floorUninsulated                           2.999e-04  7.676e-06   39.071  < 2e-16 ***
in.insulation_foundation_wallUninsulated                -2.006e-04  1.145e-05  -17.512  < 2e-16 ***
in.insulation_foundation_wallWall R-15, Exterior        -2.129e-03  3.788e-05  -56.211  < 2e-16 ***
in.insulation_wallBrick, 12-in, 3-wythe, R-15           -3.044e-04  1.541e-05  -19.755  < 2e-16 ***
in.insulation_wallBrick, 12-in, 3-wythe, R-19           -6.869e-05  1.936e-05   -3.549 0.000387 ***
in.insulation_wallBrick, 12-in, 3-wythe, R-7            -1.257e-04  1.685e-05   -7.462 8.54e-14 ***
in.insulation_wallBrick, 12-in, 3-wythe, Uninsulated     3.102e-04  1.045e-05   29.692  < 2e-16 ***
in.insulation_wallCMU, 6-in Hollow, R-11                 3.843e-04  3.566e-05   10.776  < 2e-16 ***
in.insulation_wallCMU, 6-in Hollow, R-19                -4.427e-05  2.522e-05   -1.755 0.079220 .
in.insulation_wallCMU, 6-in Hollow, Uninsulated          3.313e-04  1.304e-05   25.404  < 2e-16 ***
in.insulation_wallWood Stud, R-11                       -1.758e-05  8.687e-06   -2.024 0.042976 *
in.insulation_wallWood Stud, R-15                       -1.125e-05  1.035e-05   -1.088 0.276687
in.insulation_wallWood Stud, R-19                       -3.046e-04  1.094e-05  -27.854  < 2e-16 ***
in.insulation_wallWood Stud, R-7                         2.045e-04  1.088e-05   18.789  < 2e-16 ***
in.insulation_wallWood Stud, Uninsulated                 7.372e-05  9.135e-06    8.070 7.08e-16 ***
in.ducts10% Leakage, R-4                                -1.256e-03  2.723e-05  -46.140  < 2e-16 ***
in.ducts10% Leakage, R-6                                -6.856e-04  2.875e-05  -23.845  < 2e-16 ***
in.ducts10% Leakage, R-8                                -1.303e-03  2.821e-05  -46.187  < 2e-16 ***
in.ducts10% Leakage, Uninsulated                        -1.189e-03  2.570e-05  -46.264  < 2e-16 ***
in.ducts20% Leakage, R-4                                -1.132e-03  2.702e-05  -41.880  < 2e-16 ***
in.ducts20% Leakage, R-6                                -9.894e-04  2.626e-05  -37.671  < 2e-16 ***
in.ducts20% Leakage, R-8                                -1.401e-03  2.788e-05  -50.266  < 2e-16 ***
in.ducts20% Leakage, Uninsulated                        -9.804e-04  2.567e-05  -38.198  < 2e-16 ***
in.ducts30% Leakage, R-4                                -1.097e-03  2.739e-05  -40.063  < 2e-16 ***
in.ducts30% Leakage, R-6                                -9.132e-04  2.810e-05  -32.501  < 2e-16 ***
in.ducts30% Leakage, R-8                                -1.424e-03  2.802e-05  -50.842  < 2e-16 ***
in.ducts30% Leakage, Uninsulated                        -9.997e-04  2.581e-05  -38.740  < 2e-16 ***
in.ductsNone                                            -1.279e-03  3.019e-05  -42.350  < 2e-16 ***
date_time                                                1.341e-11  1.927e-12    6.961 3.40e-12 ***
`Dry Bulb Temperature [°C]`                              9.207e-05  7.848e-07  117.318  < 2e-16 ***
`Relative Humidity [%]`                                 -2.202e-06  1.653e-07  -13.317  < 2e-16 ***
`Wind Speed [m/s]`                                       2.830e-05  9.550e-07   29.630  < 2e-16 ***
`Wind Direction [Deg]`                                  -8.237e-10  1.525e-08   -0.054 0.956933
`Global Horizontal Radiation [W/m2]`                    -1.049e-06  2.024e-08  -51.798  < 2e-16 ***
`Direct Normal Radiation [W/m2]`                         7.309e-07  1.612e-08   45.335  < 2e-16 ***
`Diffuse Horizontal Radiation [W/m2]`                    9.680e-07  2.742e-08   35.298  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.0006184 on 187416 degrees of freedom
Multiple R-squared:  0.4529,    Adjusted R-squared:  0.4527
F-statistic:  2185 on 71 and 187416 DF,  p-value: < 2.2e-16
```

Other models we tried include a KSVM model that we made using the cart and kernlab packages. After creating training and testing sets and running the ksvm() function on usage_per_sqft, we found that the KSVM has a training error of about 0.18 and an accuracy of about 0.0005, so it's safe to say we will not be using a KSVM model for this one.

```
Support Vector Machine object of class "ksvm"

SV type: eps-svr  (regression)
 parameter : epsilon = 0.1  cost C = 5

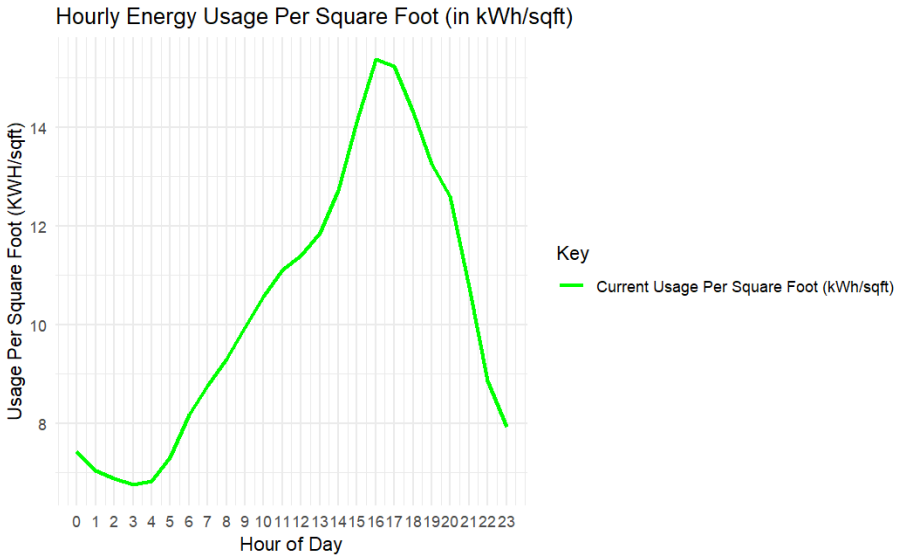
Gaussian Radial Basis kernel function.
 Hyperparameter : sigma =  0.0373887858790331

Number of Support Vectors : 5590

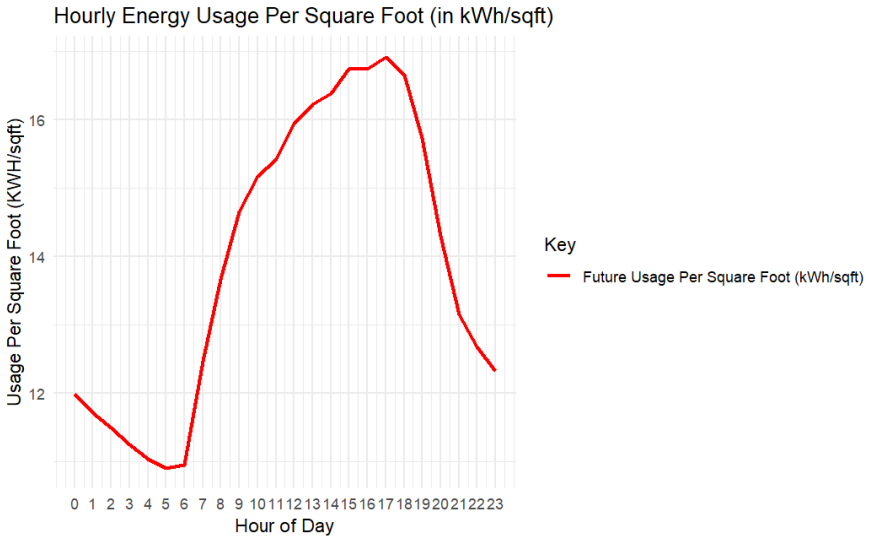
Objective Function Value : -7136.061
Training error : 0.181176
Cross validation error : 0
Laplace distr. width : 0.001152
```

We then created a brand new dataframe, hotter_data, by making a copy of modelDF and then adding 5°C to each value in the temperature column. By using our linear model, we decided to run the predict() function on hotter_data to create a new column, hotter_usage_pred, that consists of the predicted energy usage per square foot provided the temperature increases by 5°C. We also extracted the hour from the date/time column and promptly removed said column after doing so for the purpose of graphing by the hour.

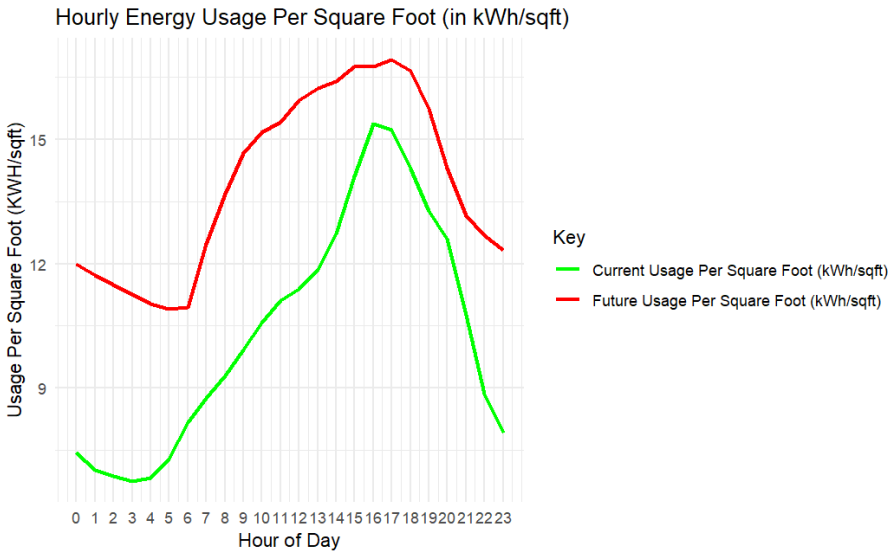
We then created another dataframe, hourly_usage, to use for creating models in ggplot. We sorted everything by hour and created two values that would be graphed—current_sum and hotter_sum. The current sum is the sum of all values in the usage_per_sqft column, while the hotter sum is the sum of all values in the hotter_usage_pred column. Using a line plot (from the ggplot package), we graphed the current total energy usage per square foot on an average 24-hour day.



The current peak usage occurs at hour 16:00 (4PM) and is about 15.4kWh/sqft. It dips a bit early on into the morning (12-3AM) before it slowly starts to increase from 3-4AM. Then, when people start to wake up (presumably to get ready for their jobs), it starts to increase very sharply from 4AM throughout the next 12 hours, peaking at 4PM. It decreases a little bit for the next hour before it starts to decrease sharply at 5PM. It continues to decrease throughout the night as the temperature starts to cool down. We then graphed the predicted future total energy usage on a typical 24-hour day, also using a ggplot line plot.



The predicted future peak usage occurs at hour 17:00 (5PM) and is about 16.9kWh/sqft. It seems the usage at midnight starts relatively higher than how it is currently, then decreases a little more steadily until 6AM. Here, it increases at an exponential rate, slowing down a little bit throughout the afternoon but still increasing. Then, after 5PM, it doesn't decrease too much over the next hour, but after 6PM, it begins to decrease much more rapidly. It doesn't slow down until 9PM, but it still continues to dip into the night. Right off the bat, both the increase and decrease in energy usage are much steeper. But in order to see how this truly matters, we'll need to graph it against the predicted future total energy usage per square foot across a typical 24-hour day. So that's precisely what we did, and the results were quite staggering.



As you can see, the predicted total energy usage is much higher than the current total energy usage. After subtracting the future peak from the current peak, then dividing that value by the current peak, we found that this would be a 10% increase in peak energy usage over the course of just one year. This is a concerningly large increase in demand, especially given that energy usage typically increases by around 2-3% each year.

Approach 2

We began the second approach in the same way as the first. We used the same feature engineering strategy and converted our building characteristics into ordered numeric scales which kept the inherent efficiency ratings. The higher the rating the better it would be. However, for the “heat pump” specifically, we online that it is the most efficient so we placed that as the highest within “cooling_order”. We placed values such as “None” or “Uninsulated” at the bottom. In parallel, we also developed categories for ease of use when visualizing the data such as “size_category” and “age_category” which are based on set ranges we determined due to the range of in.sqft being between 633-8194 with a large majority being in the “Small” to “Medium” category. (Approach 2, Lines 200-370)

After the base methodology was created, we began creating two different types of models in this approach. Model 1 outlines a basic linear regression model while Model 2 outlines a linear regression model with interaction modelling.

Model 1: Linear Regression Model

Similar to how we began with “Approach 1” we developed a baseline linear model called “lmModel” predicting "usage_per_sqft" using weather features (temperature, humidity, solar radiation), time features (hour, day of week), and house characteristics (insulation levels, window type, duct configuration, HVAC efficiency, bedrooms, stories, and occupants). This gave us an adjusted R-squared of approximately 0.252.

We iteratively refined this model by removing non-significant predictors, creating “lmModel_base_only_sig” with an adjusted R-squared of approximately 0.251. This model only included values that were significant in “lmModel”, removing attributes such as "day_of_week". The next model we developed was called, "lmModel_insulation" which focused exclusively on weather and insulation variables. We believed that this would be an excellent approach as insulation traps heat and cold, therefore the higher the insulation, the less energy is needed. However, the data did not agree with this sentiment as the adjusted R-squared values was approximately 0.105. The final model we tried with this approach was "lmModel_house_size" which we added occupancy and structural features to. This showed more promise then the “lmModel_insulation” but did not beat the other two models with an adjusted R-squared value of 0.179.

Finally when we compared all the models, we found the baseline model performed the best at an adjusted R-squared value of 0.2520. (Approach 2, Lines 425-524)

Description: df [4 × 4]

Model <chr>	R_Squared <dbl>	Adj_R_Squared <dbl>	Num_Predictors <dbl>
lmModel	0.2521282	0.2520404	22
lmModel_base_only_sig	0.2519533	0.2518695	21
lmModel_house_size	0.1799293	0.1798550	17
lmModel_insulation	0.1050250	0.1049868	8

4 rows

Model 2: Linear Regression Model with Interaction Terms

In Model 2, we took a different approach as we were trying to identify different ways that the attributes will interact with each other. For instance, we understood that a single degree increase in temperature could cause a much higher energy demand for the client then someone with a well-insulated house. However, this was only speculation, so we developed the interaction terms to test whether this was a viable hypothesis. We created various interaction terms such as “temp_×_wall_insulation” and “temp_×_ceiling_insulation” which represent the level of insulation and the impact it has when referenced to temperature. Another term is “temp_×_humidity” which models the combined effect of heat and humidity on the cooling load. A full list can be found here where we converted it into a function as it was easier to call in: (Approach 2, Lines 170-186).

```r
## Calculate interaction terms
### Give the name of the new dataframe and automatically calculate the interaction terms.
```{r}
interaction_calc = function(df){
    # Calculate interaction terms
    df$temp_x_insulation_wall = df$`Dry Bulb Temperature [°C]` * df$in.insulation_wall
    df$temp_x_insulation_ceiling = df$`Dry Bulb Temperature [°C]` * df$in.insulation_ceiling
    df$temp_x_hvac = df$`Dry Bulb Temperature [°C]` * df$in.hvac_cooling_efficiency
    df$temp_x_windows = df$`Dry Bulb Temperature [°C]` * df$in.windows
    df$temp_x_window_area = df$`Dry Bulb Temperature [°C]` * df$in.window_areas
    df$hour_x_temp = df$hour * df$`Dry Bulb Temperature [°C]`
    df$temp_x_humidity = df$`Dry Bulb Temperature [°C]` * df$`Relative Humidity [%]`
    df$solar_x_windows = df$`Global Horizontal Radiation [W/m2]` * df$in.windows
    df$solar_x_window_area = df$`Global Horizontal Radiation [W/m2]` * df$in.window_areas

    return(df)
}
```
```

For this model, we changed the prediction target to “total_usage” from “usage_per_sqft” as the interaction effects were able to be shown in absolute completion terms. We created three different linear models with interactions called: "lm_with_interactions", "lm_with_interactions_sig", "lm_with_interactions_custom". The first "lm_with_interactions" is a baseline with all the interaction terms we have with an adjusted R-squared value was approximately 0.387. The second model "lm_with_interactions_sig" was only the significant values from "lm_with_interactions" with an adjusted R-squared value was approximately 0.375. The third model "lm_with_interactions_custom" was a based on the rational of what we believed was significant based on observations, it was more of a test then a data driven model, however it did show us how ill placed our observations were as the data did not back it up. The adjusted R-squared value was approximately 0.305. (Approach 2, Lines 527-656)

Description: df [3 × 4]

Model <chr>	R_Squared <dbl>	Adj_R_Squared <dbl>	Num_Predictors <dbl>
lm_with_interactions	0.3874205	0.3871590	80
lm_with_interactions_sig	0.3757607	0.3755509	63
lm_with_interactions_custom	0.3057309	0.3055716	43

3 rows

Comparison: Model 1 vs Model 2

We found when comparing the two models that our best model came from a variation of Model 2 called “lm_with_interactions”. This significantly out performed the Model 1 variations and when ranked was first amounst the different models we tested. The interaction model “lm_with_interactions” substantially outperformed the baseline, demonstrating that building characteristics meaningfully moderate the temperature-energy relationship. A full result list is attached. (Approach 2, Lines 659-665)

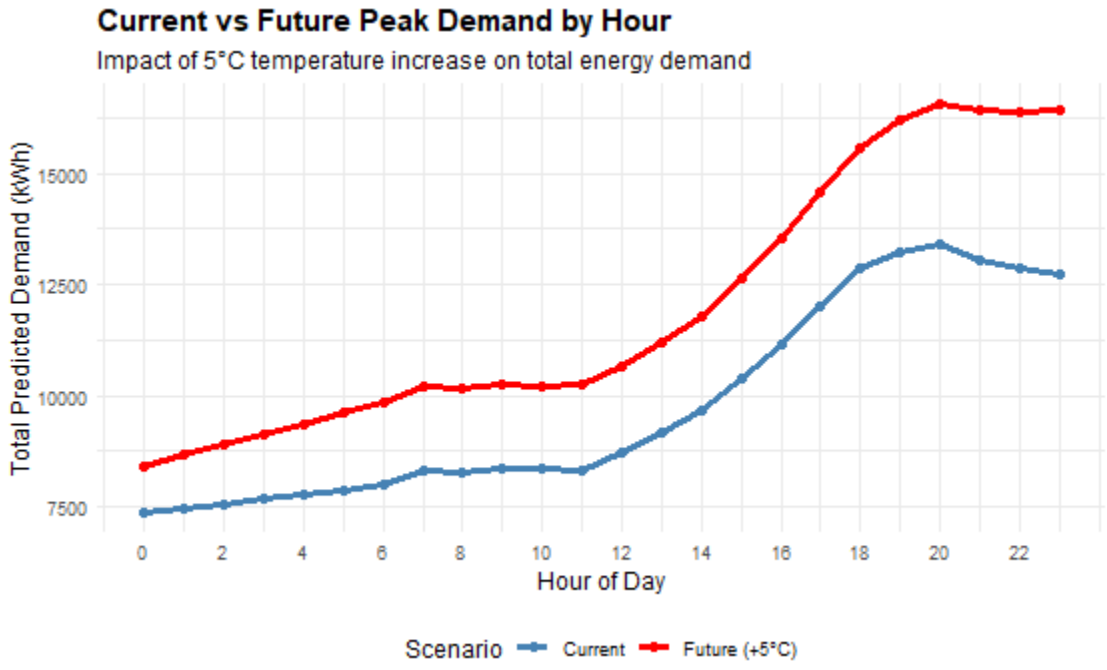
Description: df [7 x 4]

Model <chr>	R_Squared <dbl>	Adj_R_Squared <dbl>	Num_Predictors <dbl>
lm_with_interactions	0.3874205	0.3871590	80
lm_with_interactions_sig	0.3757607	0.3755509	63
lm_with_interactions_custom	0.3057309	0.3055716	43
lmModel	0.2521282	0.2520404	22
lmModel_base_only_sig	0.2519533	0.2518695	21
lmModel_house_size	0.1799293	0.1798550	17
lmModel_insulation	0.1050250	0.1049868	8

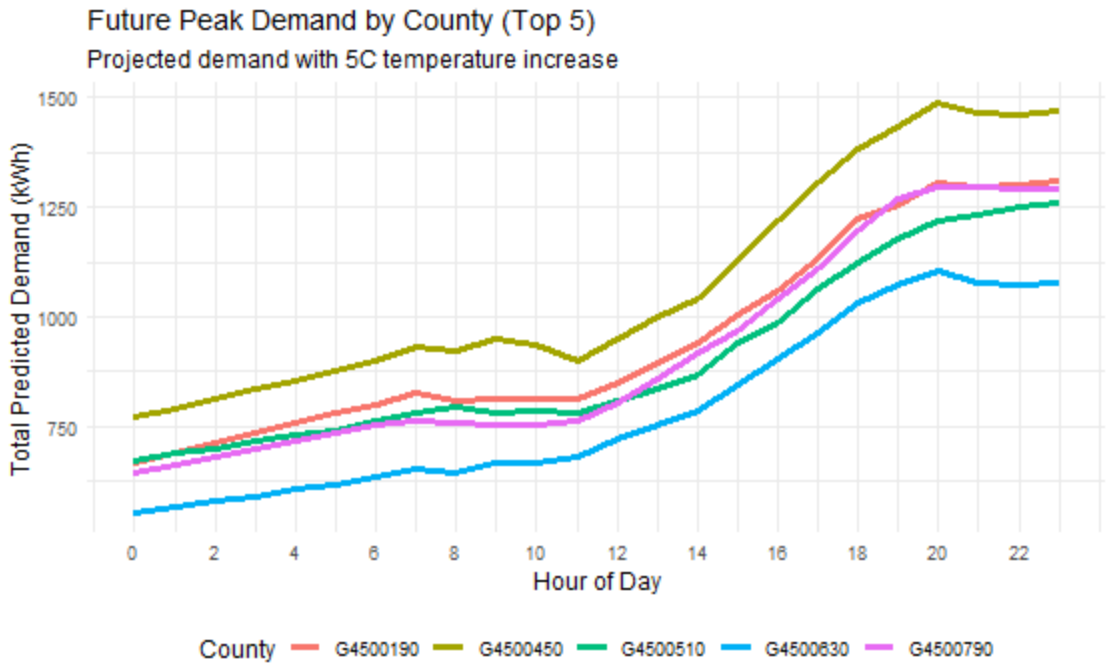
7 rows

Future Demand

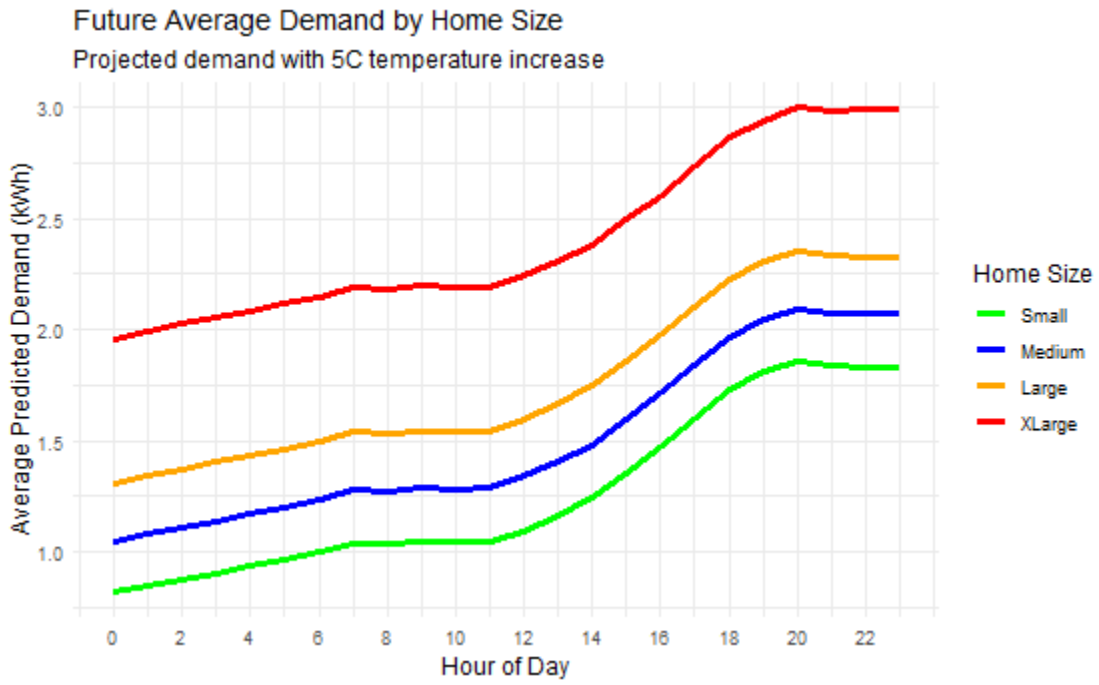
In this approach we created a future scenario dataset by adding 5°C to all temperature values and recalculating the dependent interaction terms. Using our trained model, we predicted energy consumption under this warmer scenario. We found that a 5°C increase in temperature increases the energy demand by 23.62% at the peak hour of 20:00 (8:00PM). The current total peak demand at that same time is 13389.31 kWh whereas with this temperature increase we see it rise to 16552.24 kWh, with all other factors remaining the same.

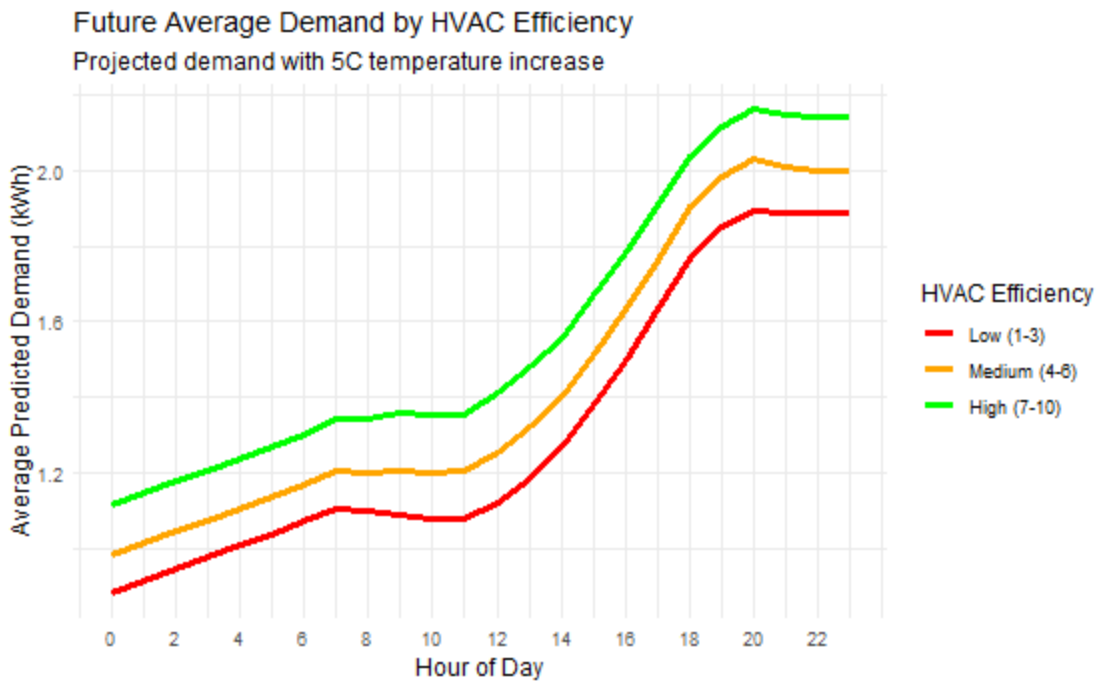


As future demand rises, by county we find the top 5 counties to be focused on are:



The demand for energy increases with the increase in temperature as well based on these findings:

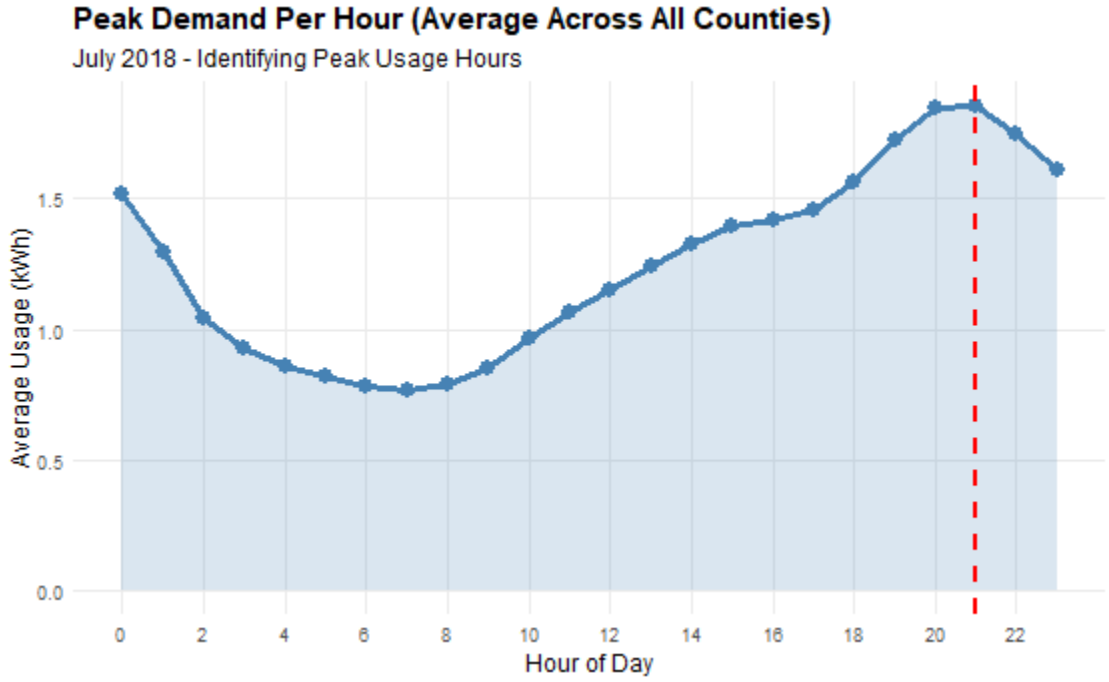


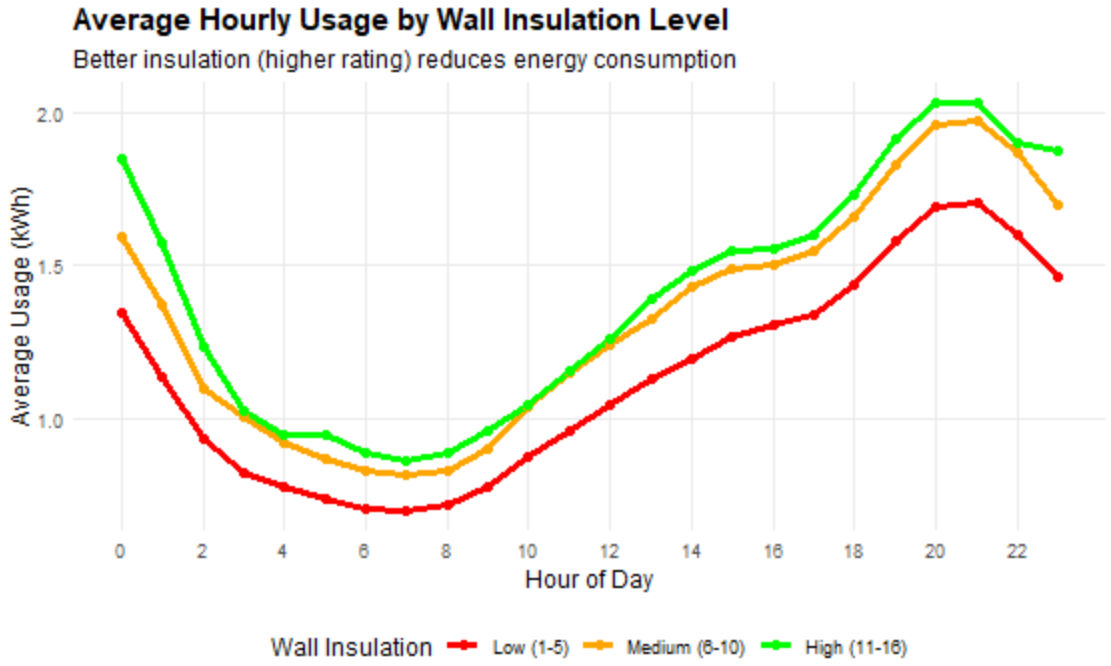


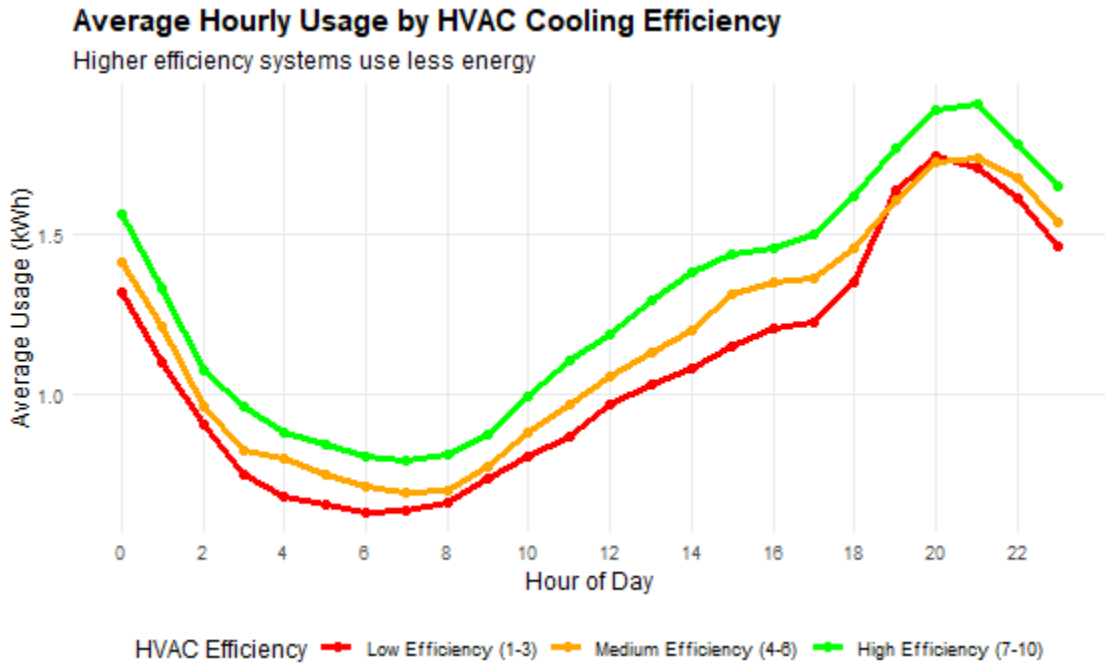
Other Findings

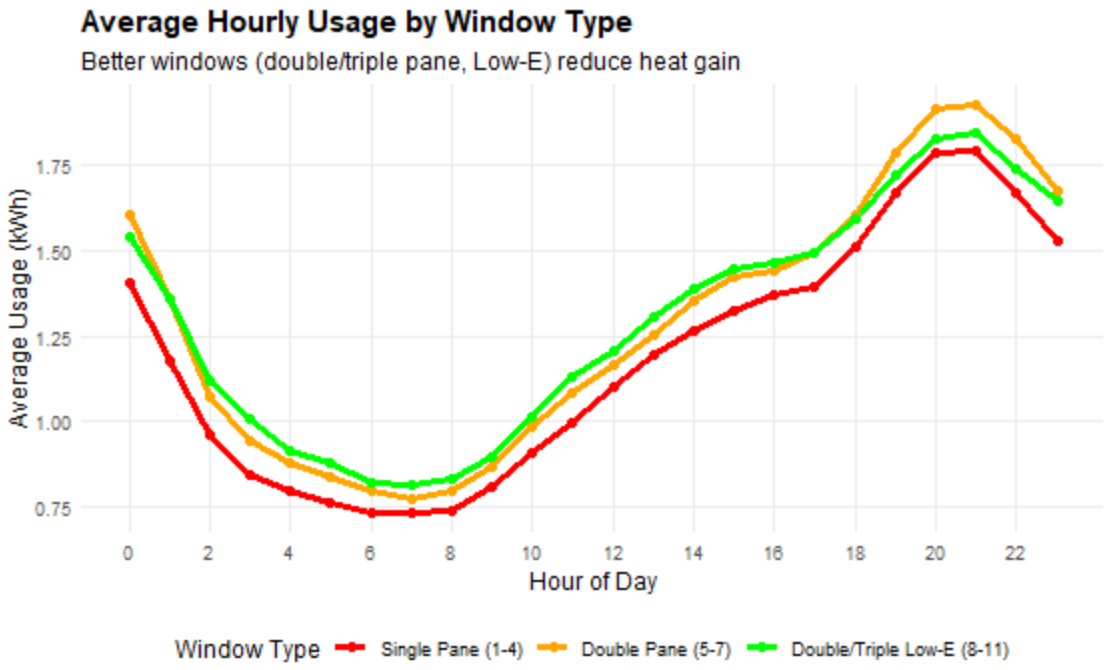
To understand the key metrics to drive business decisions we developed six different visualizations to help eSC understand where they should focus their time and energy.

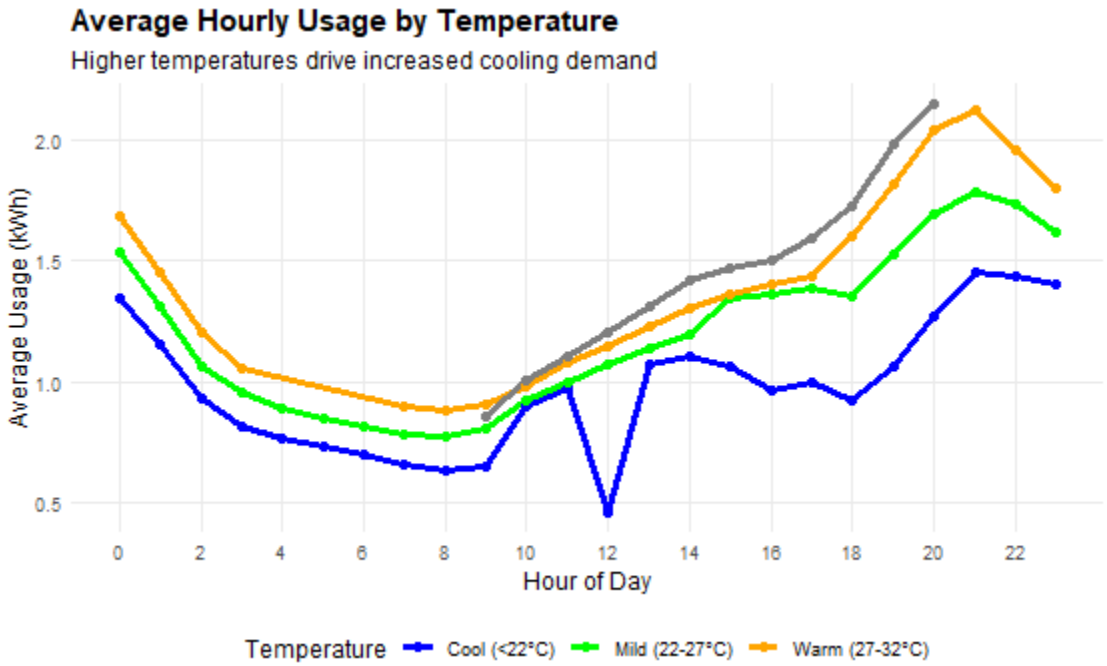
Peak Demand Per Hour, Usage by Wall Insulation Level. Usage by HVAC Efficiency, Usage by Window Type, Usage by Temperature Band, and Usage by Home Size.

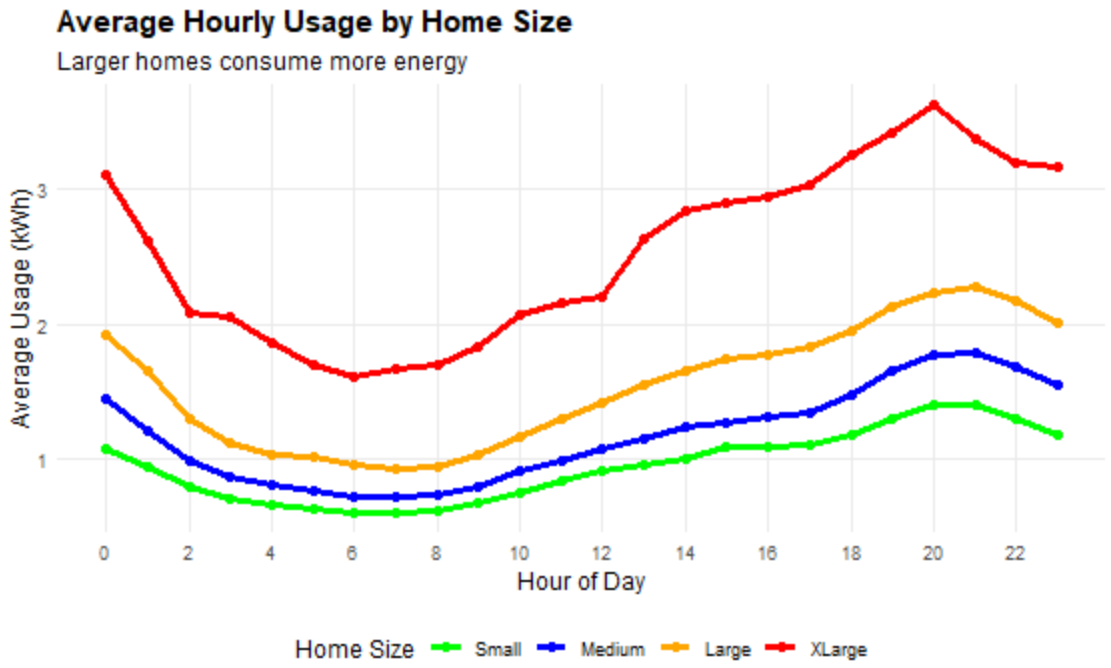



Average Hourly Usage by Wall Insulation Level
Better insulation (higher rating) reduces energy consumption
Average Usage (kWh)
2.0
1.5
1.0
0
2
4
6
8
10
12
14
16
18
20
22
Hour of Day
Wall Insulation
Low (1-5)
Medium (6-10)
High (11-16)

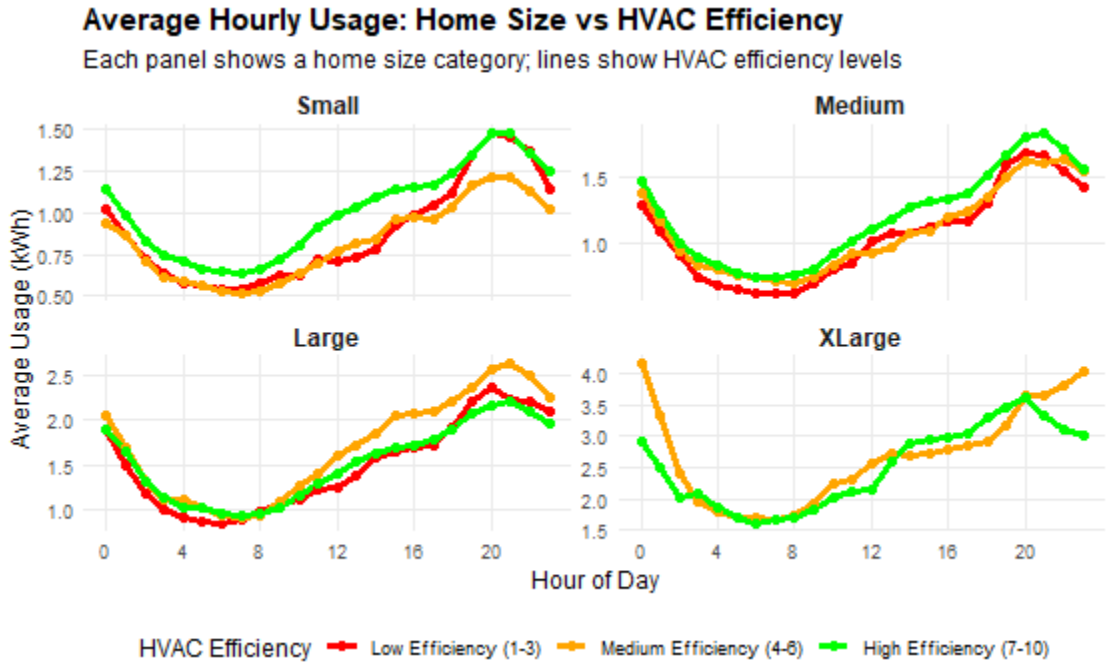

Average Hourly Usage by HVAC Cooling Efficiency
Higher efficiency systems use less energy
Average Usage (kWh)
1.5
1.0
0
2
4
6
8
10
12
14
16
18
20
22
Hour of Day
HVAC Efficiency
Low Efficiency (1-3)
Medium Efficiency (4-6)
High Efficiency (7-10)

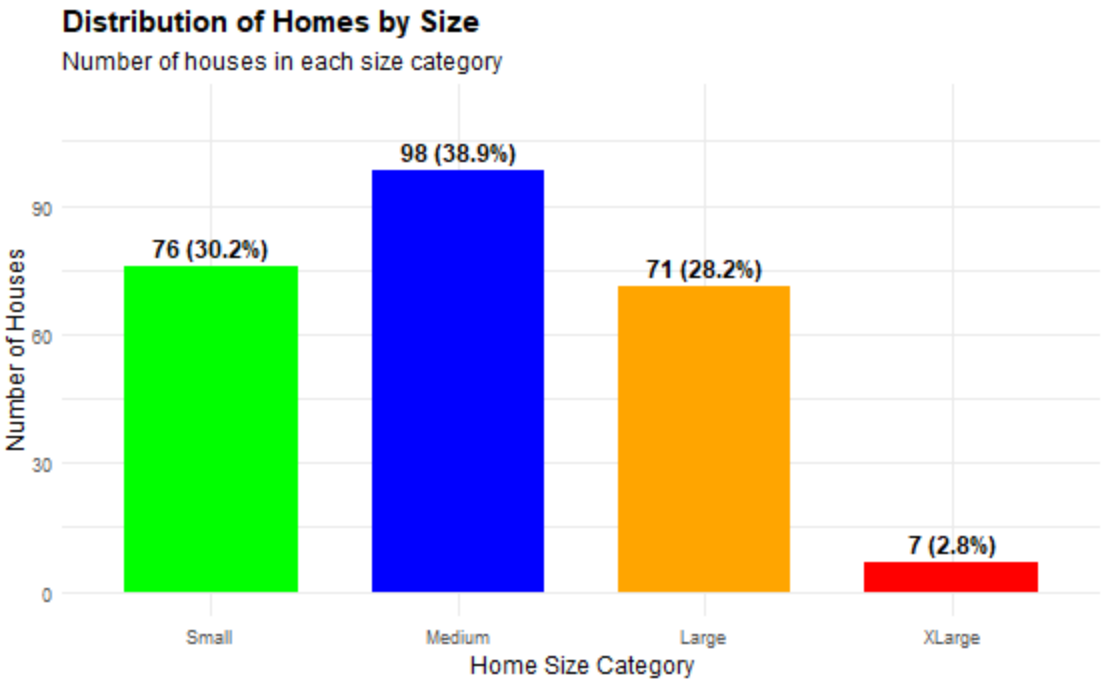

Average Hourly Usage by Window Type
Better windows (double/triple pane, Low-E) reduce heat gain
Average Usage (kWh)
1.75
1.50
1.25
1.00
0.75
0
2
4
6
8
10
12
14
16
18
20
22
Hour of Day
Window Type
Single Pane (1-4)
Double Pane (5-7)
Double/Triple Low-E (8-11)

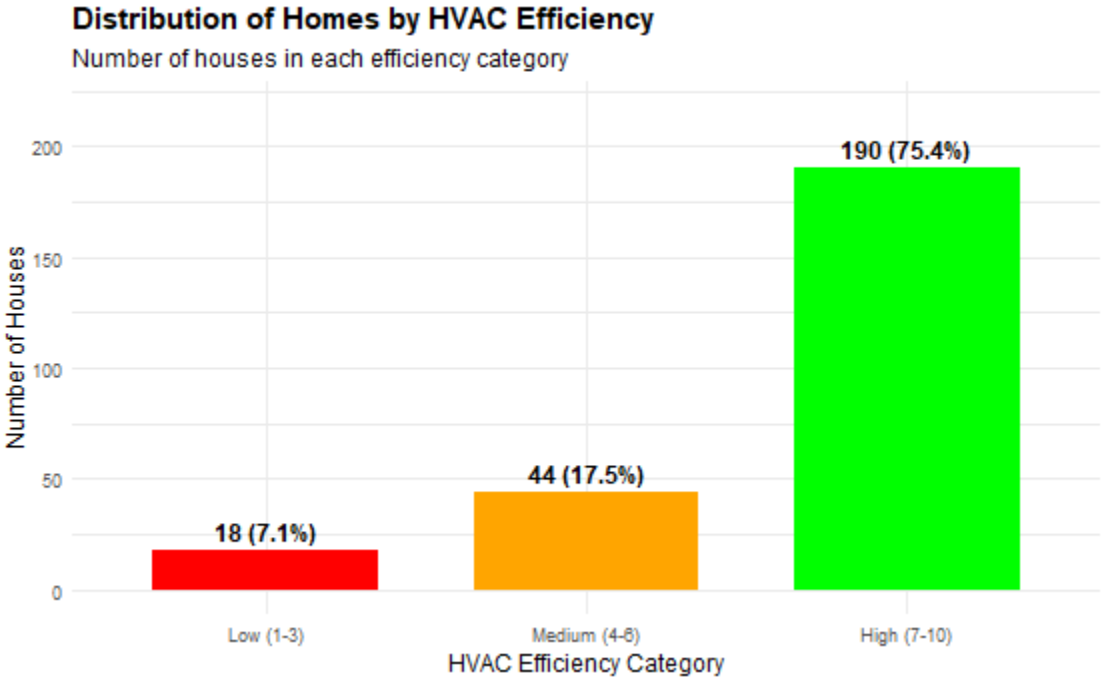

Average Hourly Usage by Temperature
Higher temperatures drive increased cooling demand
Average Usage (kWh)
2.0
1.5
1.0
0.5
0
2
4
6
8
10
12
14
16
18
20
22
Hour of Day
Temperature
Cool (<22°C)
Mild (22-27°C)
Warm (27-32°C)

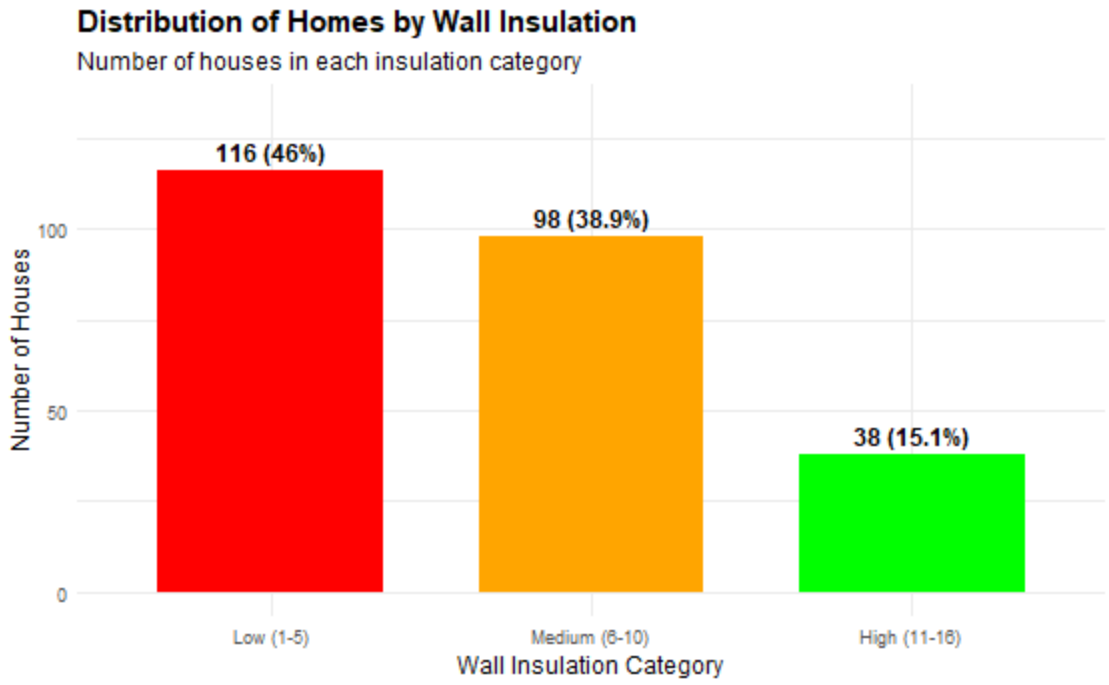


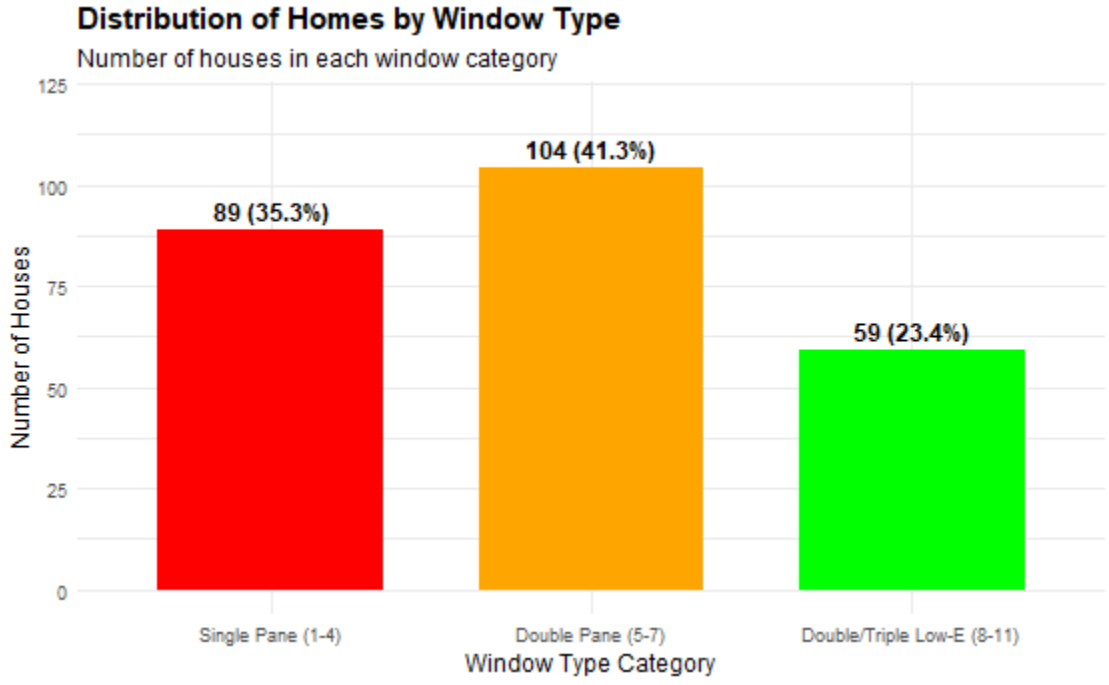
However, these do not provide enough context due to the sample size of the houses when compared to the overall sample set. For this find the following graphs provide that context.

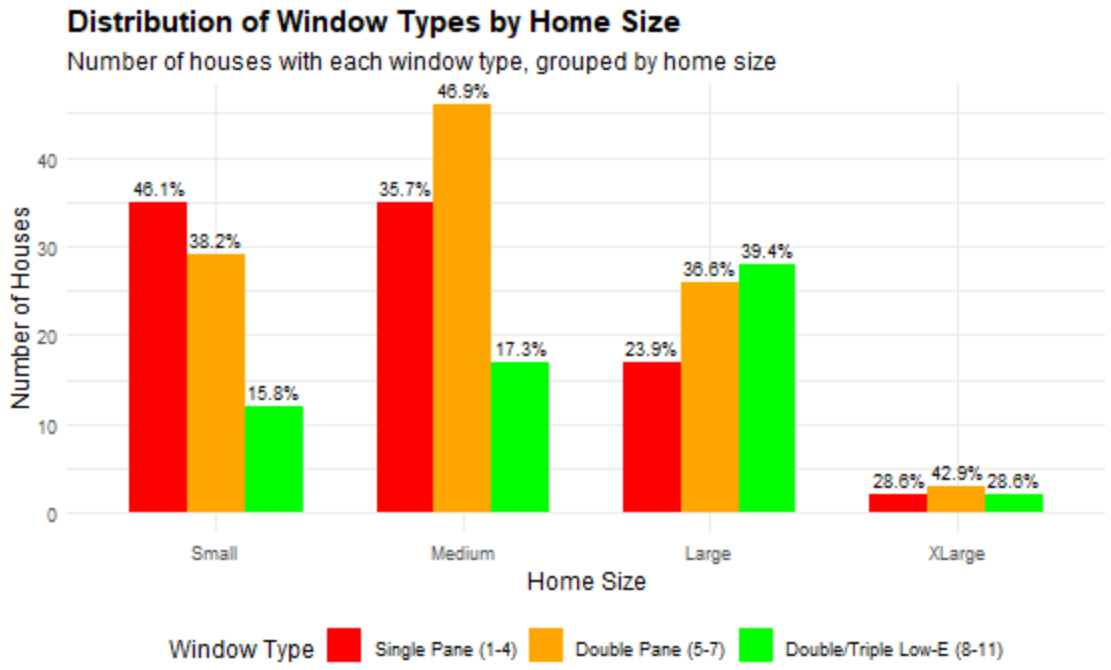


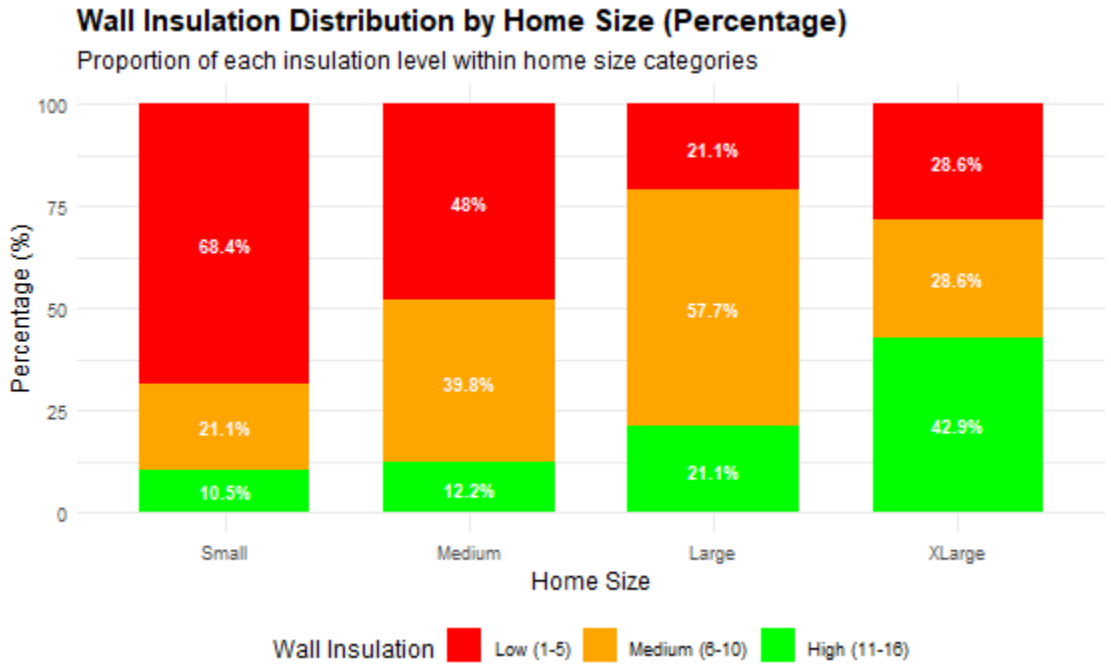

Distribution of Homes by Size
Number of houses in each size category
98 (38.9%)
76 (30.2%)
71 (28.2%)
7 (2.8%)
90
60
30
0
Number of Houses
Small
Medium
Large
XLarge
Home Size Category

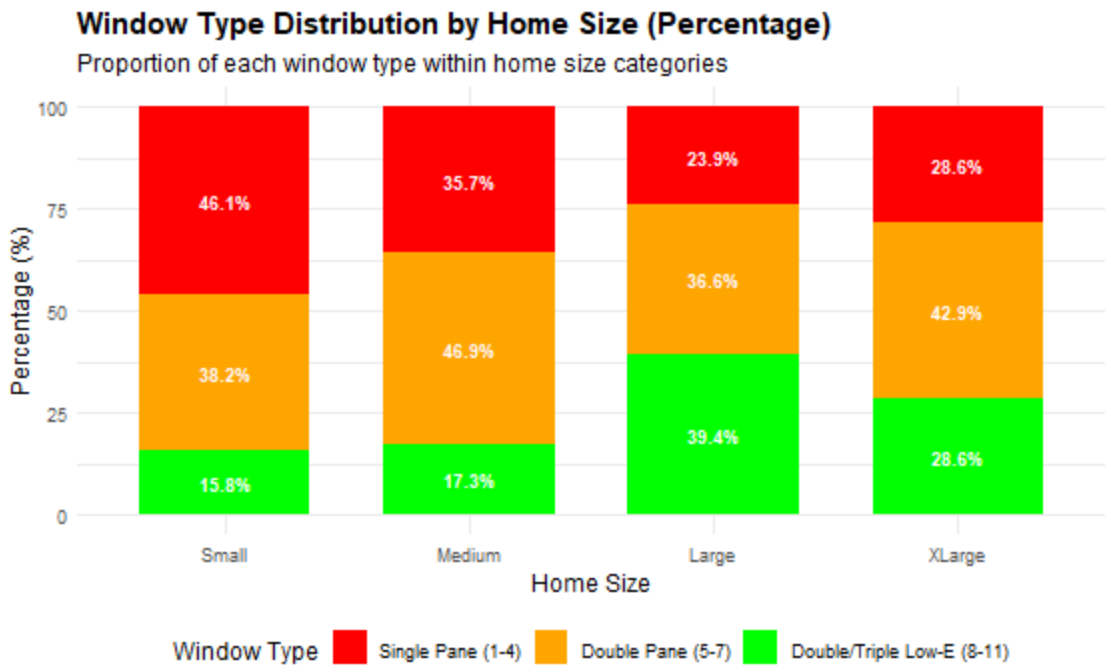

Distribution of Homes by HVAC Efficiency
Number of houses in each efficiency category
190 (75.4%)
44 (17.5%)
18 (7.1%)
200
150
100
50
0
Number of Houses
Low (1-3)
Medium (4-6)
High (7-10)
HVAC Efficiency Category


Distribution of Homes by Wall Insulation
Number of houses in each insulation category
116 (46%)
98 (38.9%)
38 (15.1%)
100
50
0
Number of Houses
Low (1-5)
Medium (6-10)
High (11-16)
Wall Insulation Category


Distribution of Homes by Window Type
Number of houses in each window category
125
100
75
50
25
0
89 (35.3%)
104 (41.3%)
59 (23.4%)
Number of Houses
Single Pane (1-4)
Double Pane (5-7)
Double/Triple Low-E (8-11)
Window Type Category


Distribution of Window Types by Home Size
Number of houses with each window type, grouped by home size
Number of Houses
0
10
20
30
40
46.1%
38.2%
15.8%
35.7%
46.9%
17.3%
23.9%
36.6%
39.4%
28.6%
42.9%
28.6%
Small
Medium
Large
XLarge
Home Size
Window Type
Single Pane (1-4)
Double Pane (5-7)
Double/Triple Low-E (8-11)


Wall Insulation Distribution by Home Size (Percentage)
Proportion of each insulation level within home size categories
Percentage (%)
0
25
50
75
100
68.4%
21.1%
10.5%
48%
39.8%
12.2%
21.1%
57.7%
21.1%
28.6%
28.6%
42.9%
Small
Medium
Large
XLarge
Home Size
Wall Insulation
Low (1-5)
Medium (6-10)
High (11-16)



Validation

Approach 1

By running our linear model, we found that it had an adjusted root mean squared error (RMSE) of around 0.0006. Our code was also able to run with no errors whatsoever. With these statistics in mind, it is safe to say our model is decently reliable in predicting energy usage per square foot. Though the adjusted R-squared was not what we had hoped for, we feel that the RMSE, alongside the fact that all NAs have been cleared and no errors were found in running the code, can be reliable enough of an indicator for our results and analysis.

Approach 2

Once we had determined that "lm_with_interactions" was the best model we could make, we began validation. We validated our best model using an 80/20 train-test split via createDataPartition, on total usage to ensure representative distributions in both sets. The model achieved a "Test RMSE: 0.6373 and Test MAE: 0.4368" (Approach 2, Lines 667-691). These results are not optimal and are barely passable as there is a average predication error of 0.6373kWh and absolute deviation from actual usage of 0. 4368kWh. This showed us that there is a unknown identifier that is more aligned with the dataset to predict energy usage.

Actionable Insights

One solution could be to increase prices. Sure, customers won't like it if the energy bill is more expensive, but that will likely deter them from using so much to save money, ultimately leading to less energy demand. One other option would be to install equipment that uses less energy. Using and installing more energy efficient equipment will reduce the base load of energy used at all times, putting less of a burden on the grid, which would, in turn, reduce the energy demand. Another option would be to invest in energy storage. A battery could be put in place to charge up with energy from the grid when demand is low and then when demand is high the battery can be used to provide some electrical energy, effectively reducing the amount of power taken from the grid and thus reducing the peak energy demand. Given what we have learned throughout this analysis, we now have a much clearer idea of what drives peak energy usage and what can potentially drive peak energy usage in the future as temperature increases, and we hope these possible solutions can help make a difference should the energy demand get too high moving forward.

Alternatively, based on the findings from "Approach 2" we recommend a targeted solution to implement a window and wall insulation upgrade program specifically for small and medium-sized homes. Our analysis reveals that small and medium homes represent 69.1% of the housing population (30.2% small, 38.9% medium), making them the highest-impact segment for intervention. These homes exhibit the largest efficiency gaps: 46.1% of small homes and 35.7% of medium homes still rely on single-pane windows, compared to only 23.9% of large homes. The disparity in wall insulation is even more pronounced: 68.4% of small homes have low insulation ratings versus just 21.1% of large homes. By offering rebates or incentive programs to encourage these homeowners to upgrade to double-pane Low-E windows and improve wall insulation, eSC could significantly reduce cooling loads during peak afternoon hours when demand is highest. Unlike HVAC upgrades, where 75.4% of homes already have high-efficiency systems, window and insulation improvements target a population with substantial room for improvement. This approach addresses the root cause of excessive energy consumption rather than simply managing demand through pricing or storage. Given what we have learned throughout this analysis about how building characteristics moderate the temperature-energy relationship, focusing resources on the majority population segment with the greatest efficiency deficits offers eSC the most cost-effective path to reducing peak demand and avoiding grid stress as summer temperatures rise.